



Tagged as Ready: Building the Campus AED Fleet Readiness Ledger From a Monthly Tag Scan

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Abstract. Campus safety reporting increasingly reduces the readiness of emergency equipment to a single published figure. At Slop University, Facilities' AED Fleet Readiness Ledger aggregates a monthly compliance-walk scan of each of the University's 42 wall-mounted defibrillator cabinets into one percentage cited annually in the University's safety report. We describe the ledger's design and report its operation across a fifteen-week semester. The reported figure weights a cabinet's scan recency at 90% and its pad-expiry status, drawn from a static procurement table rather than the scan itself, at 10%. We find that removing the pad-expiry term entirely changes the campus-wide Readiness percentage by well under a percentage point, a difference we characterise as not significant; Facilities retains the term for completeness. Scan-recency data show a small minority of cabinets regularly exceed the ledger's forty-five-day compliance window, with limited consequence for the headline figure given its current weighting. We discuss the Readiness percentage as an instrument that, by construction, mostly certifies that the monthly walk occurred rather than that a given cabinet would function if opened, and note the gap this leaves between what the figure reports and what it is read to mean.

Introduction

Campus safety reporting increasingly reduces the readiness of emergency equipment to a single published figure, cited alongside other assurance metrics in an institution's annual report. Wall-mounted automated external defibrillators (AEDs) are a common instance: cabinets are simple to install and difficult to continuously verify, so institutions typically substitute a periodic compliance check for a genuine functional test. Independent inspection programmes elsewhere have found that a meaningful share of public AEDs are not, in fact, ready for immediate use when checked directly, with expired electrodes and restricted access among the most common causes [1]. What such inspections make visible is not always what an institution's own reporting captures.

At Slop University, Facilities formalised its existing paper-tag compliance walk into a single instrument: the AED Fleet Readiness Ledger, which aggregates a monthly scan of each of the

University's wall-mounted defibrillator cabinets into one percentage, published annually in the University's safety report. The ledger's design choices — what a scan is taken to certify, and how a cabinet's pad-expiry status enters the roll-up — determine what the published figure can and cannot be read to mean. Administrative measurement more broadly has a documented tendency to settle on whichever quantity is easiest to collect regularly, whether or not that quantity tracks the underlying state of interest [2], [3].

This paper makes three contributions, each hedged appropriately for a single-institution case study. First, we describe the design of the AED Fleet Readiness Ledger, to our knowledge the University's first campus-wide instrument of its kind. Second, we report descriptive scan-recency patterns across a full semester of the ledger's operation over the University's 42 wall-mounted units. Third, we evaluate the sensitivity of the published Readiness percentage to its

pad-expiry weighting term, and find that the term’s removal changes the headline figure by an amount within normal reporting variance; our results suggest the current weighting may be conservative rather than load-bearing.

The remainder of the paper describes the ledger’s aggregation logic (Method), reports scan-recency patterns and the weighting ablation (Results), and considers what the published figure does and does not warrant (Discussion).

Related work

Administrative and clinical settings both document a persistent gap between what a compliance record shows and what it is taken to mean. In healthcare, targets and their associated performance figures are well known to be gamed at the point of collection rather than at the point of the underlying activity [2], a pattern that recurs even in less politically visible settings such as departmental spending classification [3]. Surgical safety checklists show a related divergence: national compliance figures can report near-universal completion while direct observation finds compliance markedly lower, and the checklist record itself predicts postoperative outcomes only weakly once observed adherence is accounted for [4], [5]. Corporate sustainability reporting exhibits a structurally similar gap between disclosed and substantiated practice [6].

A parallel literature treats the measurement instrument itself as the object of scrutiny. Citizen-science and wearable-device studies find that an easily collected proxy signal — a submitted observation, a device’s step count — correlates with the state of interest only moderately, and unevenly across settings [7], [8], [9]. Sensing infrastructure that relies on a periodic read rather than continuous ground truth faces analogous problems of drift, gap-filling, and uncertain provenance, documented across smart water metering [10], building occupancy counting [11], and Internet-of-Things calibration scheduling more broadly [12]; standard approaches to filling the resulting gaps in longitudinal covariate data carry their own, well-characterised biases [13].

Within the University, prior work has traced a comparable dynamic in the Indicator Commons’ own sunset-scoring instrument, whose operational uptime entered the same dashboard it was built to audit [14], and in the campus’s hand-sanitiser dispensers, where a refill log stood in

for a hygiene-compliance signal it tracked only weakly [15].

Method

Facilities operates 42 wall-mounted AED cabinets across eleven buildings, grouped in this paper into six zones (teaching buildings pooled, student residences, the sports centre, the library, administration, and a residual “other” category covering smaller standalone sites). Each cabinet carries a QR scan-tag, introduced to replace the University’s previous paper initialling sheet. A monthly compliance walk, conducted by a rotating roster of Facilities staff, scans each tag with a handheld reader; the scan timestamps to a central log. Cabinets not reached during a given month’s walk carry forward their previous timestamp rather than being marked missing, a convention chosen for continuity of the published figure rather than for its statistical properties.

A second data source, updated separately, is a static procurement table recording each cabinet’s electrode-pad model and printed expiry date; a pad is flagged current if the expiry date has not yet passed as of the reporting month, and lapsed or unknown otherwise. This table is refreshed annually by Facilities’ asset team and is not linked to the monthly scan event — a cabinet can be scanned on schedule with a lapsed pad, or scanned late with a current one.

The ledger combines the two sources into a single campus-wide Readiness percentage:

$$R = \frac{100}{N} \sum_{i=1}^N (\alpha \cdot \mathbb{1}[s_i \leq \tau] + (1 - \alpha) \cdot e_i)$$

where $N = 42$ is the cabinet count, s_i is the number of days since cabinet i ’s last logged scan, $\tau = 45$ days is the compliance window, $e_i \in \{0, 1\}$ is the pad-currency indicator, and α is the scan term’s weight, currently fixed by Facilities policy at:

```
readiness:
  alpha: 0.9      # weight on scan recency
  tau_days: 45   # compliance window
  expiry_source: procurement_table_v3 #
  refreshed annually, not scan-linked
```

The published figure is R computed monthly and averaged across the reporting semester. Results evaluates R ’s sensitivity to α by recomputing it under two counterfactual weightings: $\alpha = 1.0$ (scan-only, dropping the pad-expiry term

entirely) and $\alpha = 0.5$ (expiry-weighted, doubling the pad term's influence). Confidence intervals for each variant are obtained by bootstrap resampling the 42 cabinets' monthly records 2,000 times.

Results

Across the fifteen-week semester, the median days-since-scan across all 42 cabinets was 19 (IQR 9–34). Six cabinets (14%) exceeded the ledger's 45-day compliance window at least once during the semester; Figure 1 breaks this down by building zone. Student residences and the sports centre — both harder for the rostered walk to access on a fixed weekday schedule — account for three of the six, though no zone was entirely free of a lapse.

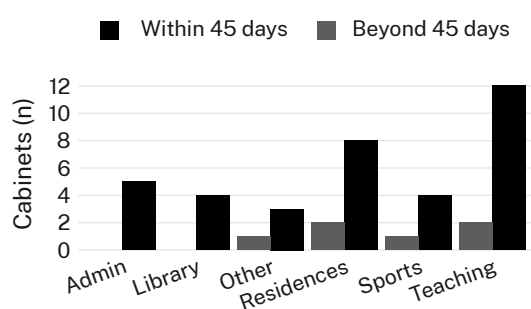


Figure 1: Six of 42 cabinets (14%) fell outside the ledger's 45-day compliance window at some point in the semester, concentrated in student residences and the sports centre.

Of the six cabinets that lapsed on scan recency, four also carried a lapsed-or-unknown pad-expiry flag in the procurement table for at least part of the semester. Under the current formula, however, the pad-expiry term contributes only 10% of the campus-wide figure ($\alpha = 0.9$), so this overlap moves the reported Readiness percentage only slightly.

Parameter	Value	Source
Compliance window (τ)	45 days	Facilities policy
Scan-recency weight (α)	0.9	Facilities policy
Pad-expiry weight ($1 - \alpha$)	0.1	Facilities policy
Reporting cadence	Monthly	Ledger schedule

We recomputed the campus-wide Readiness percentage under the current formula and two counterfactual weightings, bootstrap-resampling the semester's monthly cabinet records 2,000 times per variant. Figure 2 shows the resulting distributions. The current formula ($\alpha = 0.9$) yields a mean Readiness of 96.2% (95% CI 95.7–96.7); dropping the pad-expiry term entirely ($\alpha = 1.0$, scan-only) raises this to 96.8% (96.3–97.3), a difference of 0.6 percentage points; doubling the pad-expiry term's weight ($\alpha = 0.5$) lowers it to 95.4% (94.8–95.9), a difference of 0.85 percentage points. All three intervals overlap substantially.

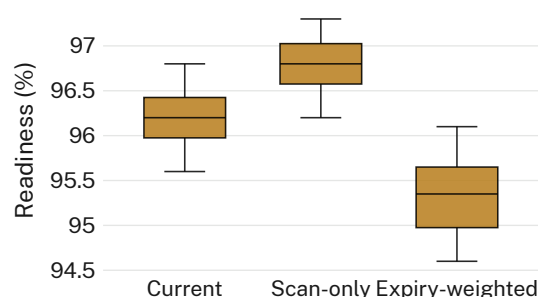


Figure 2: Removing the pad-expiry term from the Readiness formula changes the published campus-wide figure by less than one percentage point (n.s.); Facilities retains the term for completeness.

Consistent with prior instrumentation work at the University, the aggregation logic proved substantially more sensitive to its own weighting choices than to the field it was ostensibly designed to incorporate [14].

Discussion

These results suggest the AED Fleet Readiness Ledger functions, in its current form, primarily as an assurance instrument for the occurrence of the monthly compliance walk rather than as a proxy for whether a given cabinet would deliver a shock if opened. The scan-recency term dominates the published figure by construction, and the one field that speaks most directly to functional readiness — pad currency — enters at a tenth of its weight and on an update cycle decoupled from the scan itself. This is not necessarily the wrong design: a monthly walk is cheaper to sustain than a continuous telemetry feed, and a published figure that moves only within its historical range may be easier for a non-specialist reader to interpret consistently across semesters. Our results suggest, however,

that a reader encountering “96.2% Ready” in the University’s safety report would reasonably infer something closer to functional readiness than the figure, by construction, actually encodes. Facilities’ own internal documentation does not make this distinction explicit, a gap consistent with similar findings elsewhere in the University’s instrumentation programme [15].

Limitations

Our analysis covers the ledger’s aggregation logic and its recorded inputs; it does not extend to whether a given cabinet would deliver a functional shock if opened, which was outside this study’s ethical and practical scope. The formula’s demonstrated insensitivity to the one field that would move closest to that question, however, suggests the omission changes less about what the published figure already represents than it might first appear. The six-zone grouping reflects Facilities’ own devolved maintenance rosters rather than a stratified sampling design; the consistency of the scan-recency pattern across zones suggests the finding generalises across the estate, though we did not test this formally. A single semester of ledger operation limits our ability to characterise seasonal variation in scan compliance.

Conclusion

We described the design of Slop University’s AED Fleet Readiness Ledger and evaluated the sensitivity of its published Readiness percentage to the one input field most plausibly tied to functional readiness. Removing that field changed the headline figure by under a percentage point, well within the range the ledger already reports across an ordinary semester. The figure appears robust primarily because it was built to be: a scan-dominated formula reports a scan-dominated quantity, consistently. Future work will extend the ledger to ingest manufacturer battery-life telemetry directly, pending a data-sharing agreement with Facilities’ asset-management vendor, and will examine whether a comparable weighting insensitivity holds in the University’s other compliance-scan instruments.

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